

Task Class

Representation of a job that needs doing with an optional due date. Tasks can be toggled complete, prioritised and have notes attached. Where applicable, a Task's overdue status can be queried.

Definition

```
public class Task
```

- Namespace
 - TaskManagements.Models

Fields

Name	Description
Notes (string)	Descriptive notes about the Task.
DateCreated (DateTime)	The date and time when the Task was created.
TaskPriority (Priority)	The priority level of the Task. Lower value means higher priority.
DueDate (DateTime)	The due date for the Task, if any.
IsComplete (bool)	True if the Task is marked as complete.
Overdue (bool)	True if the task is overdue, false otherwise.

Methods

Name	Description
Task(string)	Construct a new Task with a description.
GetDescription()	Get the description of the Task.
SetDescription(string)	Set the description of the Task.
ToggleCompletion()	Toggle the completion state of the Task.
SaveTo(BinaryWriter)	Save the Task to a binary format.
LoadFrom(BinaryWriter)	Load the Task from a binary format.
ToString()	Return a text representation of the Task.

RepeatingTask Class

A daily or weekly repeating task.

Definition

```
public class RepeatingTask
```

- Namespace
 - TaskManagement.Models
- Inheritance
 - Task

Fields

Name	Description
Frequency (Frequency)	How often the Task repeats.
IsComplete (bool)	True if the Task is complete.

Methods

Name	Description
RepeatingTask(string, DateTime, Frequency)	Construct a new RepeatingTask with a given due date and frequency.
ToggleCompletion()	Mark an incomplete RepeatingTask as completed or a completed RepeatingTask as incomplete.
SaveTo(BinaryWriter)	Save to binary format.
LoadFrom(BinaryReader)	Load from binary format.
ToString()	Return a text representation.

Habit Class

Regularly repeating tasks that track completion streaks.

Definition

```
public class Habit
```

- Namespace
 - TaskManagement.Models
- Inheritance
 - RepeatingTask

Fields

Name	Description
Streak (int)	Number of times a Habit has been completed consecutively.

Methods

Name	Description
Habit(string, DateTime, Frequency, int)	Construct a new Habit with a given due date and expected frequency.
ToggleCompletion()	Mark an incomplete Habit as completed or a completed Habit as incomplete.
SaveTo(BinaryWriter)	Save a Habit to binary format.
LoadFrom(BinaryReader)	Load a Habit from binary format.
ToString()	Return a text representation.

Priority Struct

The priority of a Task. Lower value means a higher priority.

Definition

```
public struct Priority
```

- Namespace
 - TaskManagements.Models

Fields

Name	Description
Value (int)	The Priority's value.

Methods

Name	Description
Priority(int)	Create a Priority with the given value.
operator++(Priority)	Increment the Priority's value, thus reducing its importance.
operator--(Priority)	Decrement the Priority's value, thus increasing its importance.

TaskList Class

A collection of Tasks with a single identifying name.

Definition

```
public class TaskList
```

- Namespace
 - TaskManagements.Models

Fields

Name	Description
TotalTasksCount (int)	The total number of Tasks in the TaskList.
IncompleteTasksCount (int)	The number of incomplete Tasks in the TaskList.

Methods

Name	Description
TaskList(string)	Construct a new TaskList with a name.
GetName()	Get the name of the TaskList.
SetName(string)	Set the name of the TaskList.
AddTask(Task)	Add a Task to the TaskList.
ClearCompletedTasks()	Remove all completed Tasks from the TaskList.
SaveTo(BinaryWriter)	Save the TaskList to binary format.
LoadFrom(BinaryReader)	Load the TaskList from binary format.
ToString()	Return a text representation of the TaskList.
GetTasks()	Get a read-only list of Tasks in the TaskList.

Project Class

A collection of Tasks that do not repeat, that maintains a completion percentage.

Definition

```
public class Project
```

- Namespace
 - TaskManagements.Models

Fields

Name	Description
PercentComplete (float)	Get the percentage of completed Tasks in the Project.

Methods

Name	Description
Project(string)	Create a new Project with the given name.
AddTask(Task)	Add a Task to the Project.

TaskCollection Class

A collection of TaskLists, maintain counts of total and completed Tasks.

Definition

```
public class TaskCollection
```

- Namespace
 - TaskManagements.Models

Fields

Name	Description
TotalTasksCount (int)	The total number of Tasks in all TaskLists.
IncompleteTasksCount (int)	The number of incomplete Tasks in all TaskLists.

Methods

Name	Description
AddTaskList(TaskList)	Add a TaskList to the TaskCollection.
ClearAllCompletedTasks()	Remove all completed Tasks from all TaskLists.
Save(string)	Save the TaskCollection to a file.
Load(string)	Load the TaskCollection from a file.
PrintAllTaskListTasks()	Print all Tasks from all TaskLists.
ToString()	Return a text representation of the TaskCollection.
GetTasksSortedByName()	Get all Tasks sorted by name.
GetTasksSortedByDate()	Get all Tasks sorted by due date.
GetTasksSortedByCreationDate()	Get all Tasks sorted by creation date.
GetTasksSortedByPriority()	Get all Tasks sorted by priority.
GetHabits()	Get all Tasks that are Habits.
GetRepeatingTasks()	Get all Tasks that are RepeatingTasks.
GetDueTasks()	Get all Tasks that are due.
GetTasksWithDescription(string)	Get all Tasks with the given description.
GetAllTasks()	Get all Tasks in the TaskCollection.

StringParse Class

Static helpers for parsing friendly format Task creation strings.

Definition

```
public static class StringParse
```

- Namespace
 - TaskManagement.Helpers

Methods

Name	Description
ParseNaturalDate(string)	Get a date from a natural language string.
ParseNaturalTime(string)	Get a time of day from a natural language string.
ParsenaturalTaskCreation(string)	Get a Task with an optional due date from a natural language string.